

RAG Lab Report

Building a Knowledge Base on Amazon Bedrock for the orders_pipeline

September 10, 2026

About this document

This document explains the screenshots from the lab. The lab had two parts:

Lab: build a knowledge source from the data dictionary, the schema, the pipeline README and the quality rules. Then ask engineering questions and capture the supporting sources.

Assessment: create 10 answerable and 3 unanswerable questions. For each question, record the expected source, the actual answer, the grounding result and the failure analysis.

The AWS account ID and the user email were hidden in the screenshots.

Lab setup

AWS Region	Asia Pacific (Mumbai)
Knowledge Base name	kb-orders-pipeline
Knowledge Base ID	4N495QCGMT
Knowledge Base type	Vector store
IAM service role	Created automatically by Amazon Bedrock (AmazonBedrockExecutionRoleForKnowledgeBase_7j2cg)
Data source	Amazon S3: s3://brian-rag-system-lab/knowledge_source_samuel/
Documents	data_dictionary.md, pipeline_README.md, quality_rules.md, schema.txt
Test mode	Standard retrieval with answer generation
Answer model	Amazon Nova Micro 1.0 (APAC Nova Micro)
Source chunks per question	Up to 5 (default value)

Words used in this document

Knowledge Base	A Bedrock service that stores our documents as vectors, so a model can search them and answer questions.
Chunk	A small piece of a document. Bedrock splits each document into chunks before it saves them.
Citation	The small numbers like [1] or [2] in an answer. Each number points to the chunk that supports that part of the answer.
Grounded	The answer comes from the documents and has citations that support it.
Abstention	The model says it cannot find the information, instead of inventing an answer.

Part 1 · Building the knowledge source

Step 1 · Upload the four documents to Amazon S3

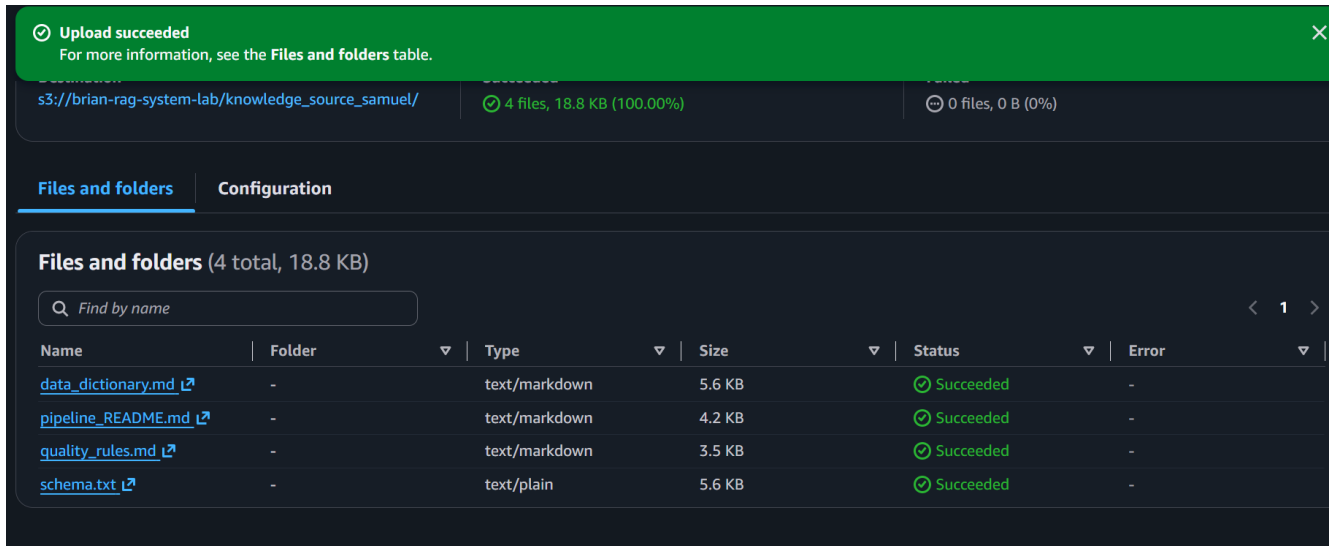


Figure 1. The four documents uploaded to Amazon S3.

The four documents were uploaded to the folder **knowledge_source_samuel** in the S3 bucket **brian-rag-system-lab**. The green message says the upload worked: 4 files (18.8 KB) succeeded and 0 files failed.

Three files are Markdown (.md). The schema was uploaded as **schema.txt** (plain text). This is important because Bedrock Knowledge Bases does not read .sql files. If the schema stayed as .sql, Bedrock would skip it and questions about the schema would fail.

Step 2 · Start creating the Knowledge Base

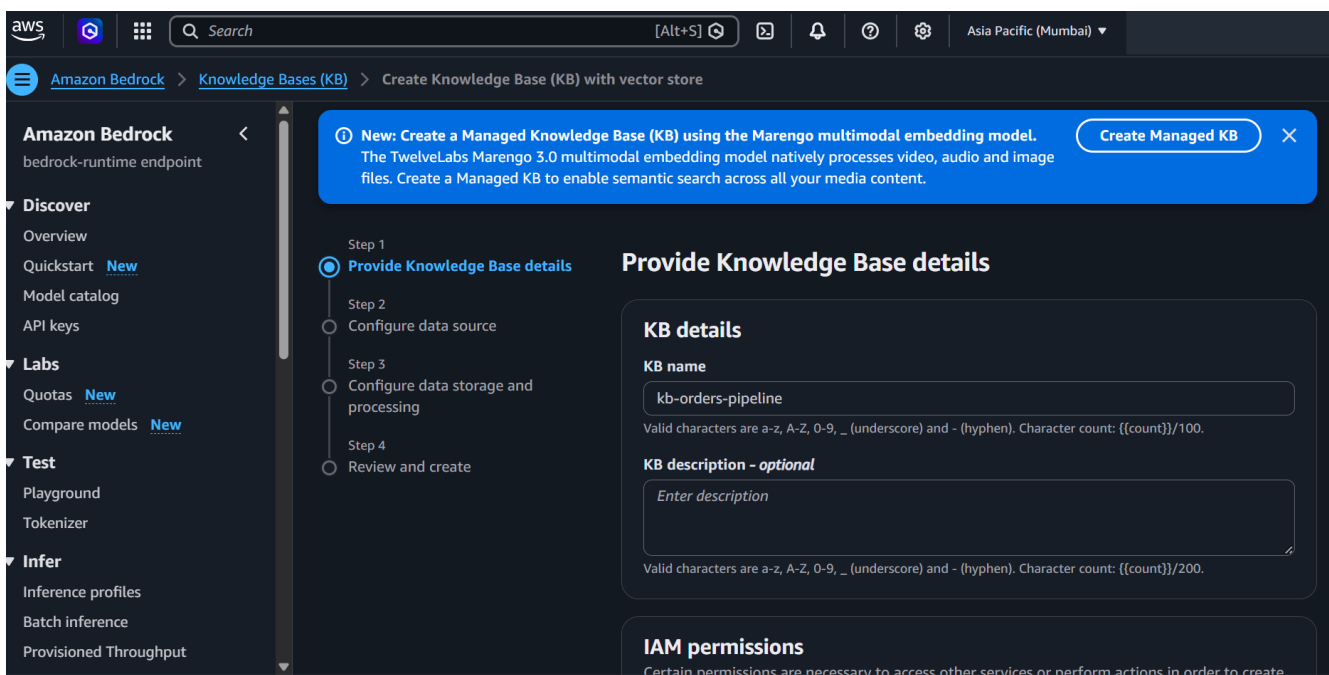


Figure 2. Step 1 of the wizard: Knowledge Base details.

In the Amazon Bedrock console, the path was **Knowledge Bases (KB) > Create Knowledge Base (KB) with vector store**. The wizard has four steps: details, data source, data storage and processing, and review. The name of the Knowledge Base is **kb-orders-pipeline**. The region is Asia Pacific (Mumbai).

The blue banner offers a different option called Managed KB. That option was not used, because this lab uses a Knowledge Base with a vector store.

Step 3 · Review and create

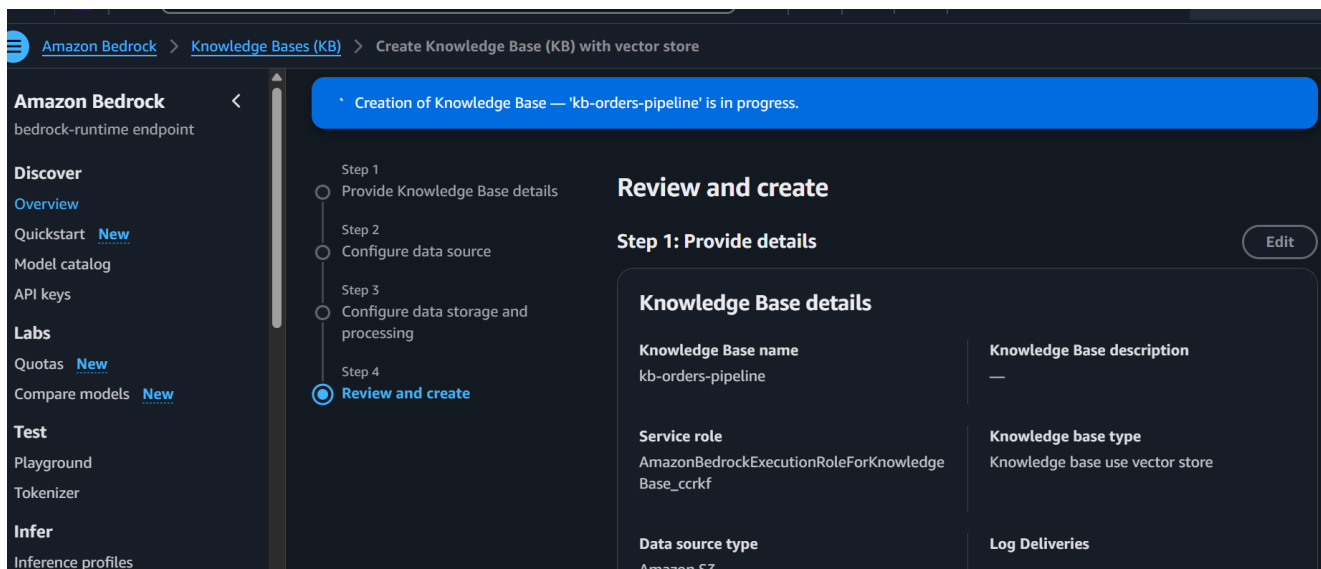


Figure 3. Step 4 of the wizard: review and create.

This is the last step of the wizard. It shows a summary before creating the Knowledge Base: the name kb-orders-pipeline, a service role created by Bedrock, the type "Knowledge base use vector store" and the data source type Amazon S3. The blue message at the top says the creation is in progress.

Step 4 · The Knowledge Base is ready

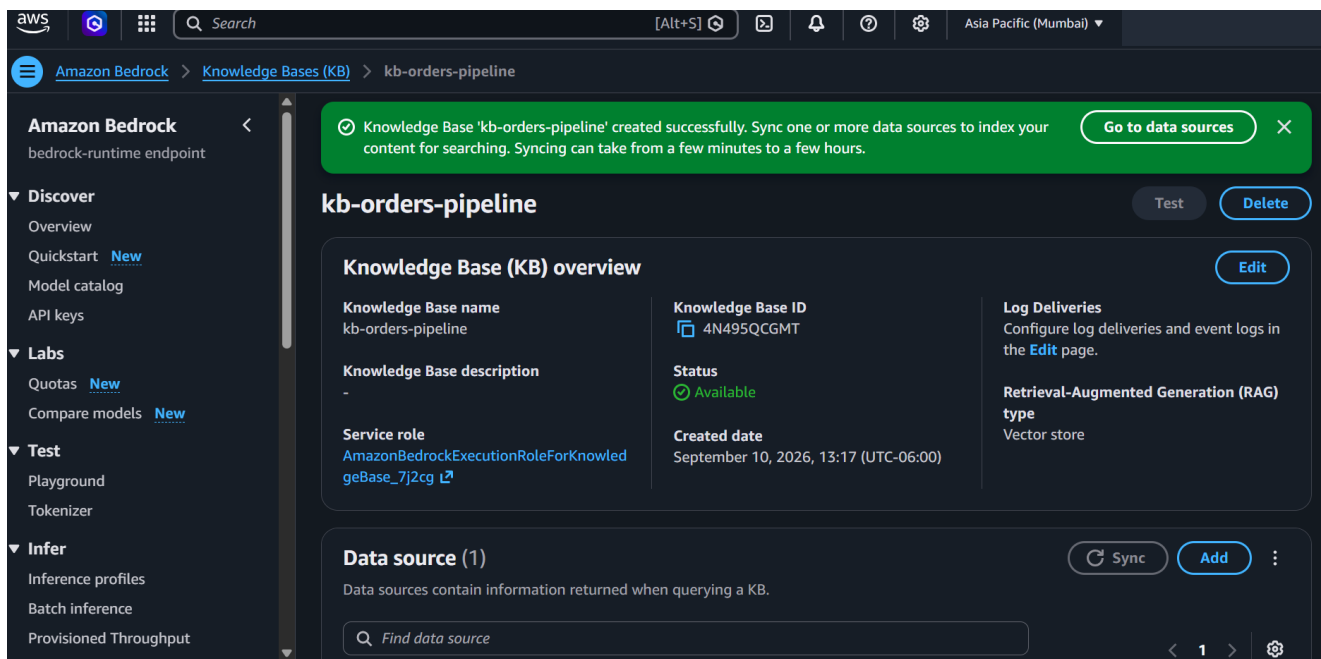
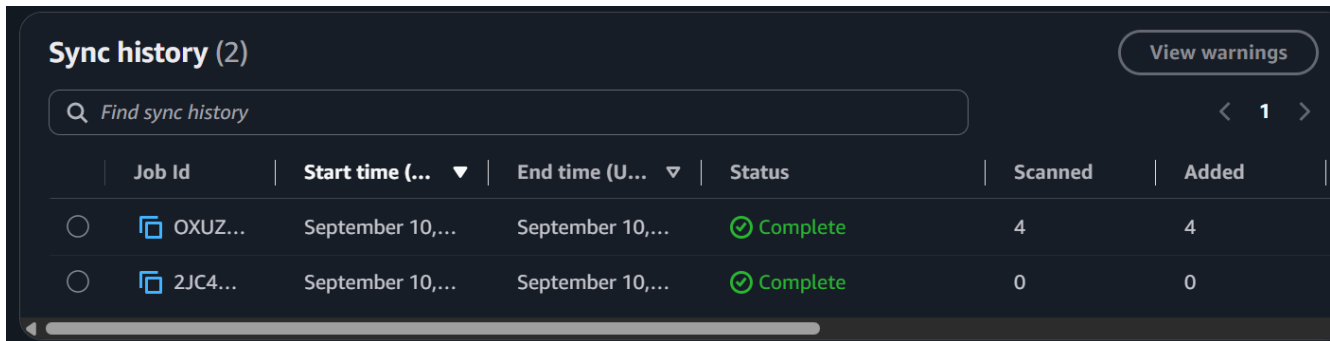


Figure 4. The Knowledge Base was created and is Available.

The green message says the Knowledge Base was created successfully. The status is **Available**, the ID is **4N495QCGMT**, and it was created on September 10, 2026 at 13:17 (UTC-06:00). The RAG type is Vector store.

The message also says that the data source must be synced. Syncing is the step that reads the files from S3, splits them into chunks, creates the vectors and saves them. The **Sync** button is in the Data source section.

Step 5 · Sync the data source



The screenshot shows a 'Sync history (2)' section with a search bar and a 'View warnings' button. Below is a table with two rows of sync jobs. The first job (2JC4...) has 0 scanned and 0 added files. The second job (OXUZ...) has 4 scanned and 4 added files. Both jobs are marked as 'Complete'.

	Job Id	Start time (... ▼)	End time (U... ▼)	Status	Scanned	Added
☐	 OXUZ...	September 10,...	September 10,...	✓ Complete	4	4
☐	 2JC4...	September 10,...	September 10,...	✓ Complete	0	0

Figure 5. Sync history of the data source.

The sync history shows two jobs. Both have the status **Complete**.

The first job (2JC4...) scanned 0 files and added 0 files. It most likely ran before the files were in the S3 folder (the Knowledge Base was created at 13:17 and the upload screenshot was taken later).

The second job (OXUZ...) scanned 4 files and added 4 files. This means all four documents were read and saved in the vector store. After this job, the Knowledge Base was ready to answer questions.

Part 2 · Engineering questions and supporting sources

All questions were asked in the **Test Knowledge Base** panel. The left side shows the settings: **Standard retrieval with answer generation** and the model **Nova Micro 1.0**. With this setting, Bedrock first searches the documents and finds the best chunks (up to 5). Then the model writes an answer using those chunks.

The small numbers in the answers ([1], [2], ...) are the citations. Each one points to the chunk that supports that sentence. The **Details** link under each answer opens the source chunks.

A01 · What data type does order_id have in staging.stg_orders, and is it the primary key?

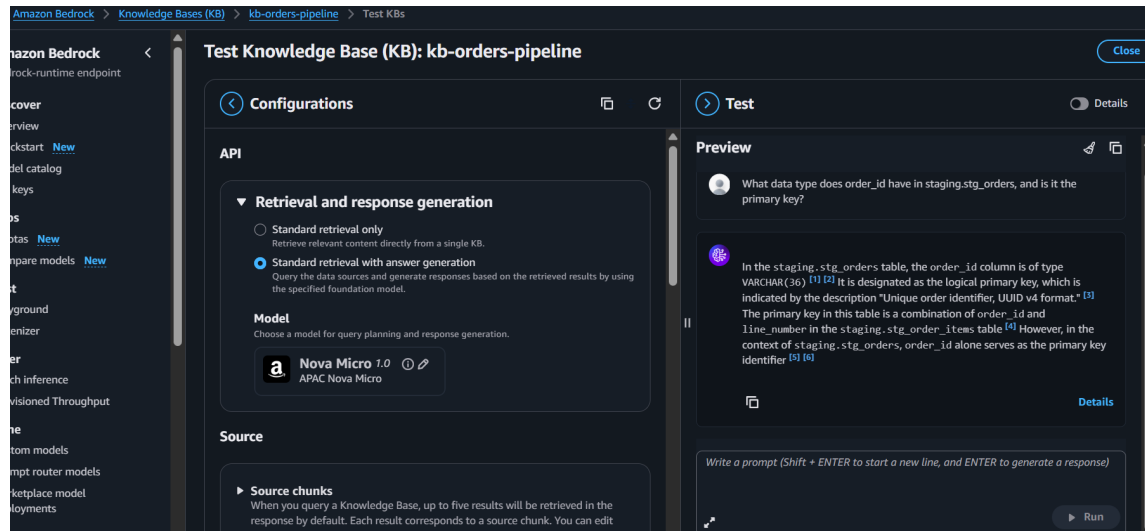


Figure 6. Test Knowledge Base panel, question A01.

Expected source: schema.txt

The model says order_id is **VARCHAR(36)** and it is the **logical primary key**, with UUID v4 format. It also adds that stg_order_items uses order_id and line_number together as its primary key. This extra part is true in the documents, but the question did not ask for it. Citations [1] to [6].

A02 · What values can the status column of an order take?

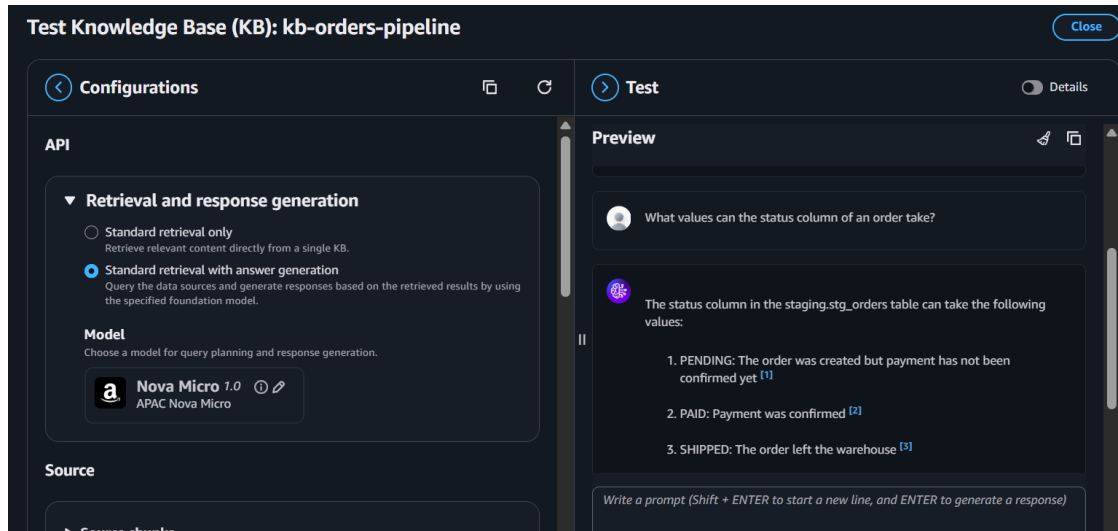


Figure 7. Test Knowledge Base panel, question A02.

Expected source: data_dictionary.md or schema.txt

The model lists the status values and explains each one: PENDING (created, payment not confirmed), PAID (payment confirmed), SHIPPED (left the warehouse), and the list continues below the visible area. The explanations come from the "Status lifecycle" part of the data dictionary. Citations [1] to [3] are visible.

A04 · Which order statuses count toward net revenue?

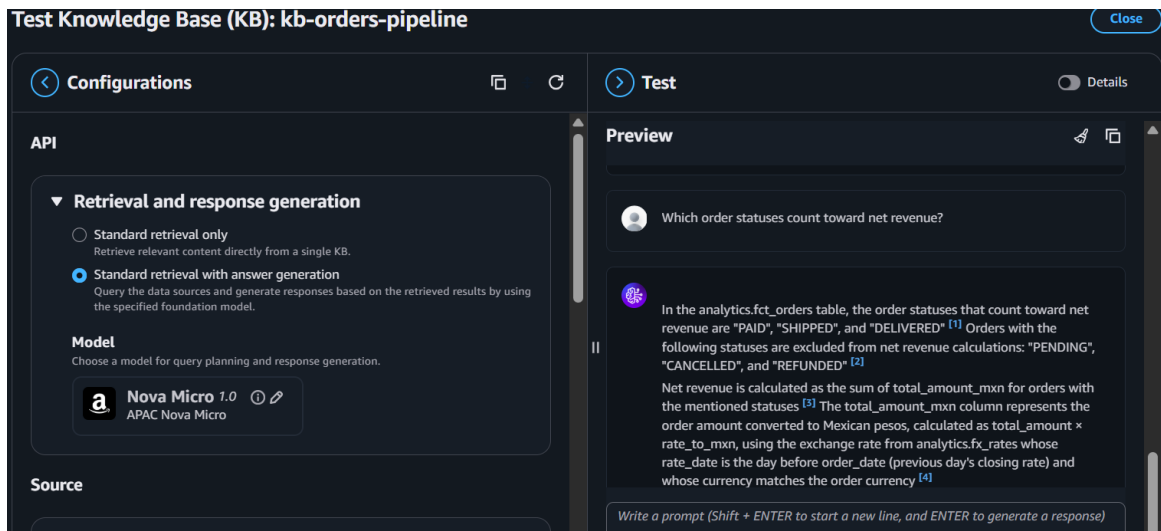


Figure 8. Test Knowledge Base panel, question A04.

Expected source: data_dictionary.md

The model says **PAID**, **SHIPPED** and **DELIVERED** count toward net revenue, and PENDING, CANCELLED and REFUNDED are excluded. It also explains how net revenue is calculated and gives the formula for total_amount_mxn (total_amount × rate_to_mxn, using the rate of the day before order_date). Citations [1] to [4].

A05 · What time does the pipeline run, and by what time must fct_orders be ready?

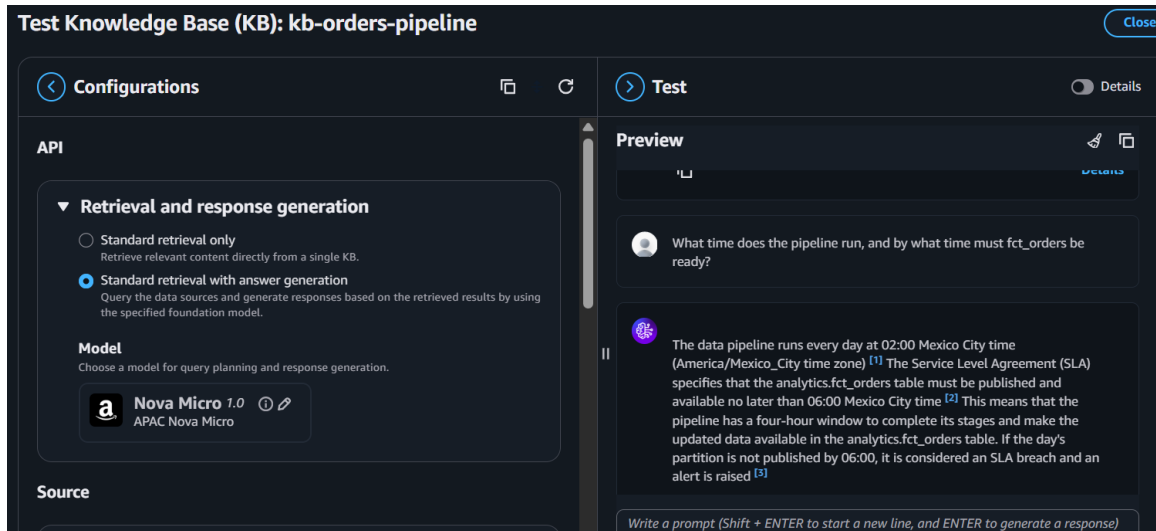


Figure 9. Test Knowledge Base panel, question A05.

Expected source: pipeline_README.md

The model says the pipeline runs every day at **02:00 Mexico City time** and the SLA says fct_orders must be ready **no later than 06:00**. It also says a late partition is an SLA breach and raises an alert. The phrase "four-hour window" is the model's own calculation from the two times; it is not written in the document. Citations [1] to [3].

A06 · I need to reprocess the orders from August 15, 2026. How do I do it?

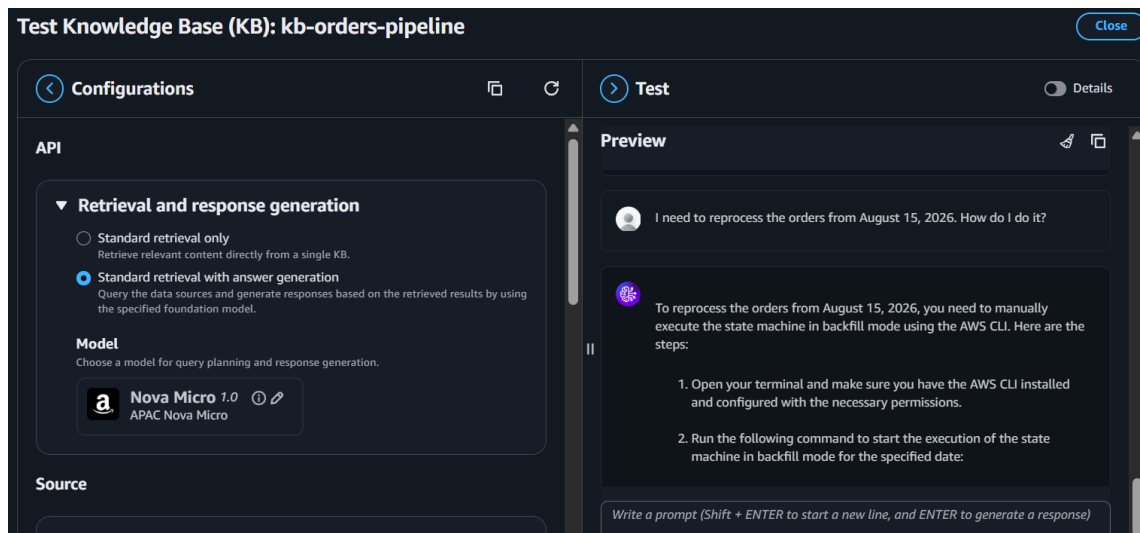


Figure 10. Test Knowledge Base panel, question A06.

Expected source: pipeline_README.md

The model says to run the state machine manually in **backfill mode** with the AWS CLI, and gives steps: open a terminal with the AWS CLI configured, then run the start-execution command for that date. The command itself is below the visible part of the screenshot.

A07 · What happens if rule DQ-001 fails?

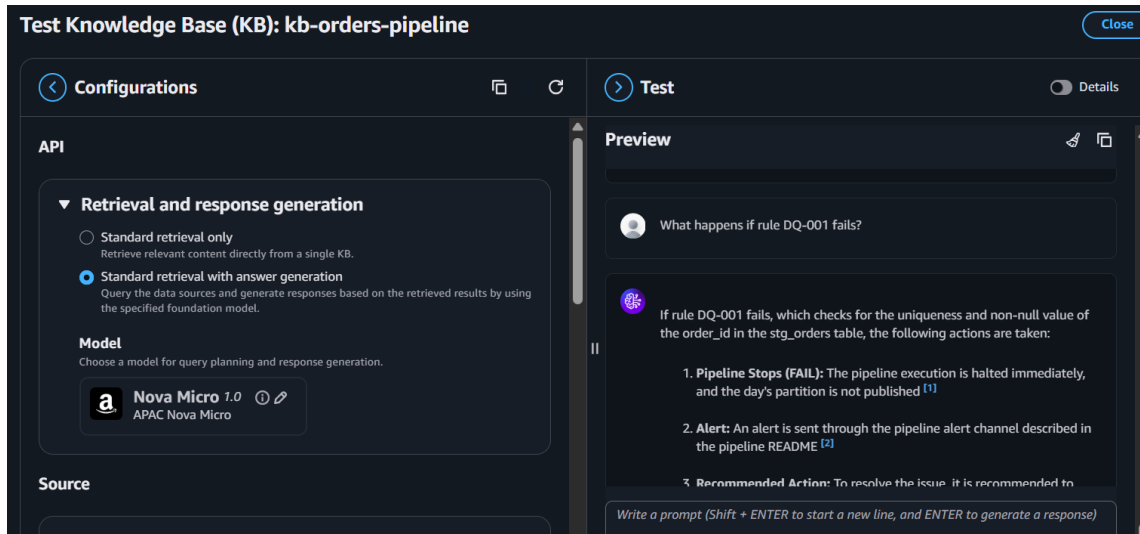


Figure 11. Test Knowledge Base panel, question A07.

Expected source: quality_rules.md

The model explains that DQ-001 checks that order_id is unique and not null. If it fails: the **pipeline stops** and the day's partition is not published [1], an **alert is sent** through the pipeline alert channel [2], and there is a recommended action to fix it (this part continues below).

A08 · At what percentage of quarantined rows does the whole pipeline fail?

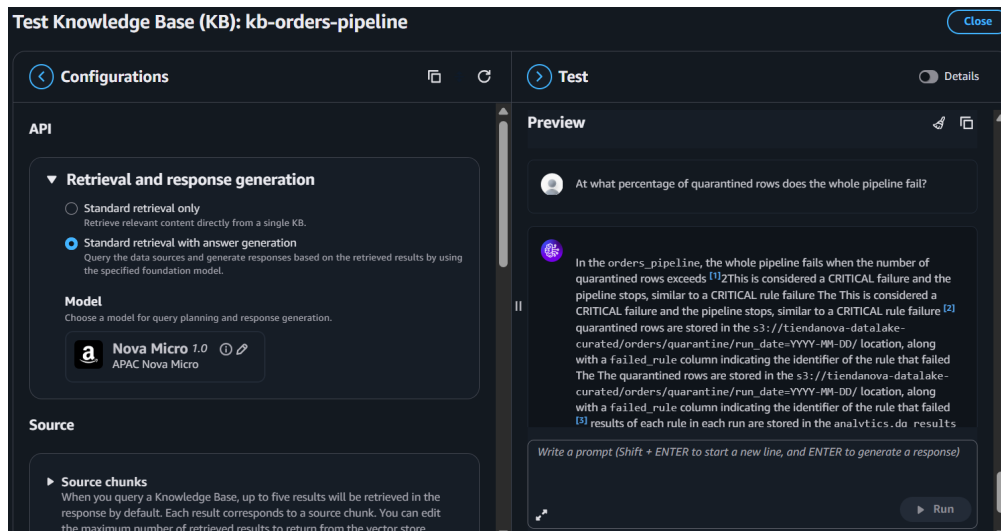


Figure 12. Test Knowledge Base panel, question A08.

Expected source: quality_rules.md

The model says the pipeline fails when quarantined rows exceed "2", but the **% sign is missing** and citation [1] appears in the middle of the number. Some sentences are **repeated two times** (the CRITICAL failure sentence and the quarantine location). It does not say that the 2% is counted over the rows of the batch from HIGH rules. The information comes from quality_rules.md, but the answer is not well written.

A09 · What type of SCD does dim_customer implement, and which columns control validity?

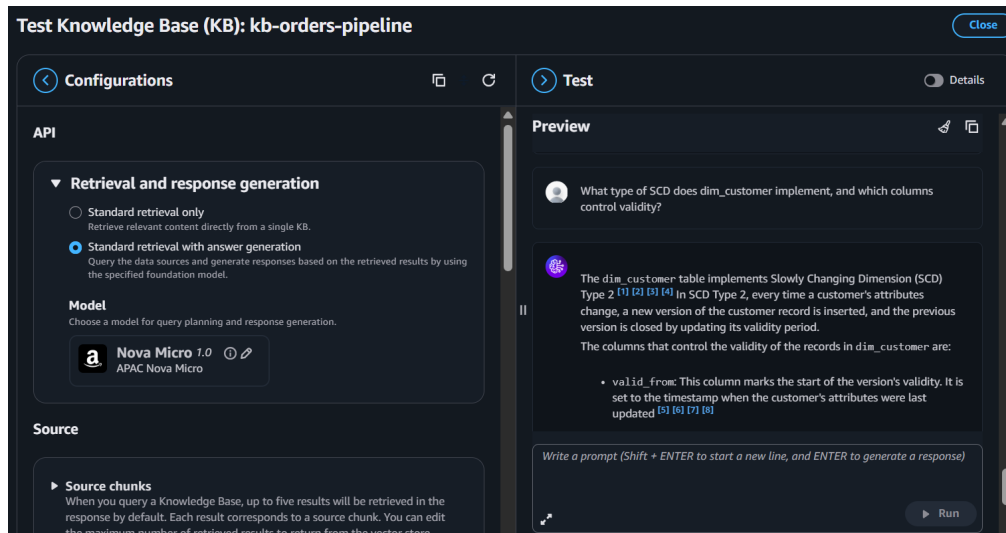


Figure 13. Test Knowledge Base panel, question A09.

Expected source: data_dictionary.md or schema.txt

The model says dim_customer is **SCD Type 2** [1] to [4] and explains that a new version is inserted when customer attributes change. It starts the list of columns with **valid_from**; the other columns continue below the visible area. The model describes valid_from with its own words ("when the attributes were last updated"); the document says "start of version validity".

A10 · If an order arrives with an invalid currency, where does that record end up and through which channel is the team notified?

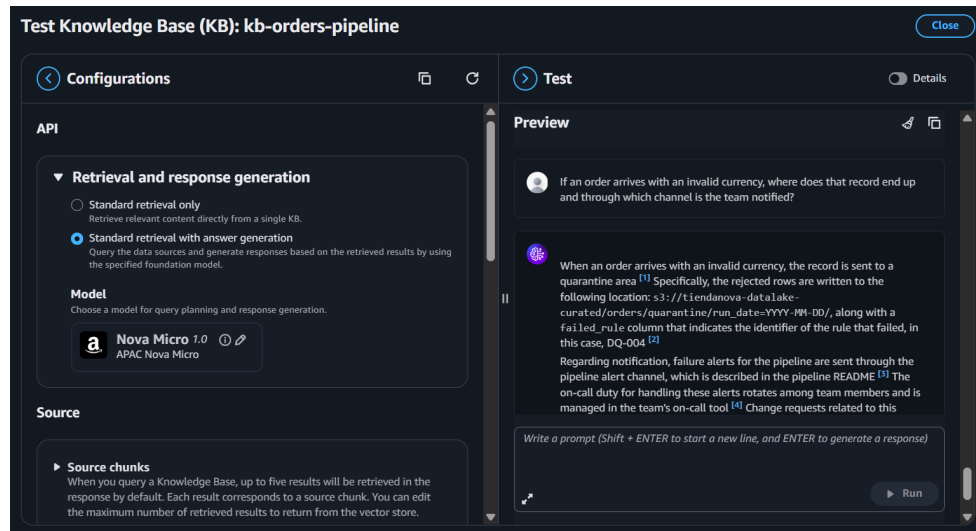


Figure 14. Test Knowledge Base panel, question A10.

Expected source: quality_rules.md and pipeline_README.md

The first part is correct: the record goes to **quarantine** in `s3://tiendanova-datalake-curated/orders/quarantine/run_date=YYYY-MM-DD/` with the `failed_rule` column set to **DQ-004** [1] [2]. The second part is weak: the model only says alerts go "through the pipeline alert channel described in the pipeline README" [3]. It does **not name the channel** (SNS topic, Slack #data-eng-alerts and the team email). Then it adds information about on-call and change requests that was not asked [4].

U01 · How much does it cost per month to run orders_pipeline on AWS?

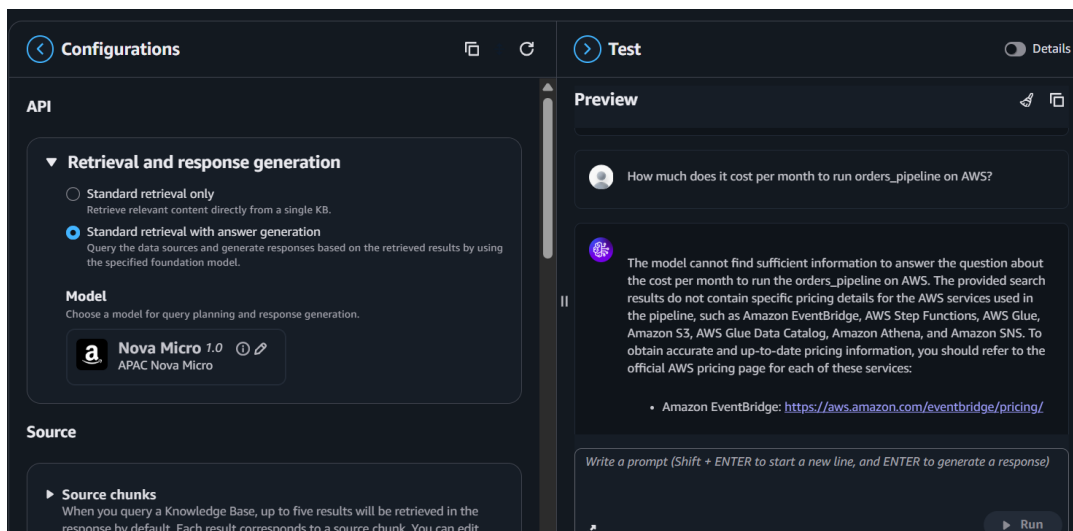


Figure 15. Test Knowledge Base panel, question U01.

Expected source: None (not in the documents)

The model says it **cannot find sufficient information** about the cost. This is a correct abstention. But after that, it lists the AWS services and gives links to AWS pricing pages (for example Amazon EventBridge pricing). These links are not in our documents; they come from the model's general knowledge.

U02 · How many days are raw-layer files kept in S3 before being deleted?

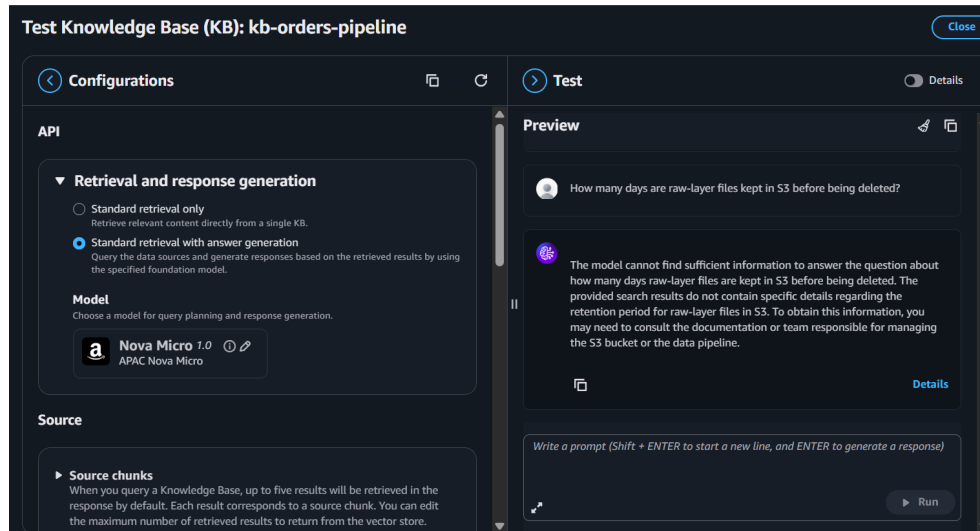


Figure 16. Test Knowledge Base panel, question U02.

Expected source: None (not in the documents)

The model says it **cannot find sufficient information** about the retention period and suggests asking the team that manages the bucket. This is a correct abstention. The model did not confuse the 3-day lookback window in the README with a retention period.

U03 · Who is the on-call person for the pipeline this week?

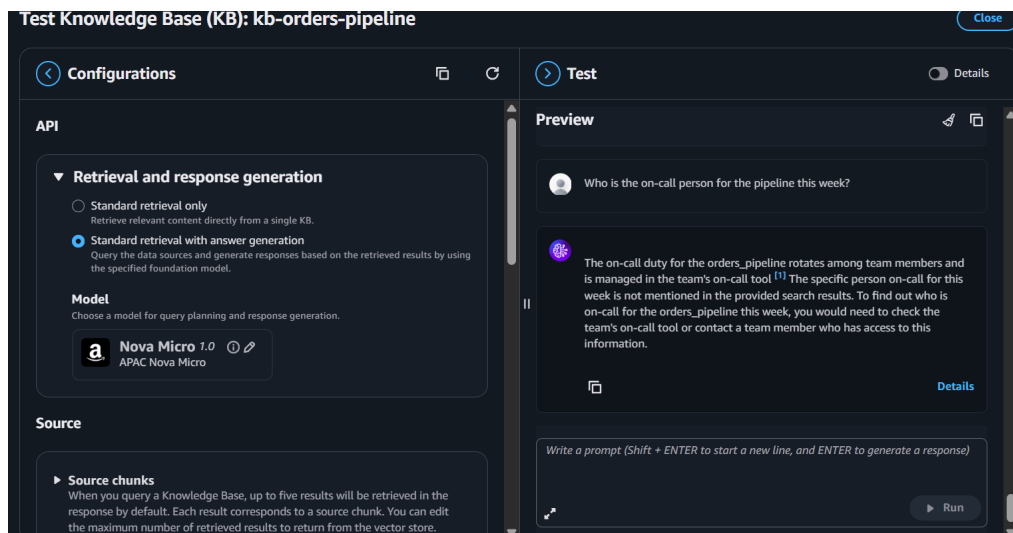


Figure 17. Test Knowledge Base panel, question U03.

Expected source: None (not in the documents)

The model says on-call duty rotates among team members and is managed in the team's on-call tool [1]. This part is true in the README. Then it says **the specific person is not mentioned** in the search results. This is a correct abstention: it did not invent a name.

Part 3 · Assessment: 10 answerable and 3 unanswerable questions

This table records, for each question, the expected source, the actual answer, the grounding result and the failure analysis. "OK" means there was no failure.

ID	Question	Type	Expected source	Actual answer (short)	Grounding result	Failure analysis
A01	What data type does order_id have in staging.stg_orders, and is it the primary key?	Answerable	schema.txt	VARCHAR(36); logical primary key; UUID v4. Also mentions the stg_order_items key.	Grounded	OK. Extra correct information that was not asked.
A02	What values can the status column of an order take?	Answerable	data_dictionary.md or schema.txt	Lists PENDING, PAID, SHIPPED, ... with the meaning of each status.	Grounded	OK.
A03	How is total_amount_mxn calculated in analytics.fct_orders?	Answerable	data_dictionary.md	total_amount × rate_to_mxn, using the fx_rates value of the day before order_date and the same currency (inside the A04 answer, Figure 8).	Grounded	OK. Not asked as a separate question; the formula appears in the A04 answer with citation [4].
A04	Which order statuses count toward net revenue?	Answerable	data_dictionary.md	PAID, SHIPPED, DELIVERED. Excludes PENDING, CANCELLED, REFUNDED.	Grounded	OK.
A05	What time does the pipeline run, and by what time must fct_orders be ready?	Answerable	pipeline_README.md	Daily at 02:00 Mexico City time; ready no later than 06:00; late partition raises an alert.	Grounded	OK. "Four-hour window" is the model's own calculation.
A06	I need to reprocess the orders from August 15, 2026. How do I do it?	Answerable	pipeline_README.md	Run the state machine manually in backfill mode with the AWS CLI.	Grounded	OK. The command is not visible in the screenshot, so the date inside it cannot be checked here.
A07	What happens if rule DQ-001 fails?	Answerable	quality_rules.md	Pipeline stops, partition not published, alert sent, recommended fix.	Grounded	OK.
A08	At what percentage of quarantined rows does the whole pipeline fail?	Answerable	quality_rules.md	"Exceeds 2" (no % sign), with repeated sentences.	Partially grounded	Generation problem. The right chunk was found, but the model broke the number, repeated text and left out "HIGH rules" and "of the batch".
A09	What type of SCD does dim_customer implement, and which columns control validity?	Answerable	data_dictionary.md or schema.txt	SCD Type 2; valid_from (list continues).	Grounded	OK. Small wording added by the model for valid_from.
A10	If an order arrives with an invalid currency, where does that record end up and through which channel is the team notified?	Answerable	quality_rules.md and pipeline_README.md	Quarantine path with failed_rule = DQ-004. Channel only described as "in the README".	Partially grounded	Multi-document gap. Part 1 (quality_rules.md) is correct. Part 2 needed the "Retries and alerts" section of the README; the model did not name SNS, Slack or email.
U01	How much does it cost per month to run orders_pipeline on AWS?	Unanswerable	None	Cannot find sufficient information; then lists AWS pricing links.	Correct abstention	Minor. The pricing links come from general knowledge, not from the documents.
U02	How many days are raw-layer files kept in S3 before being deleted?	Unanswerable	None	Cannot find sufficient information; suggests asking the team.	Correct abstention	OK. Did not confuse the 3-day lookback with retention.

ID	Question	Type	Expected source	Actual answer (short)	Grounding result	Failure analysis
U03	Who is the on-call person for the pipeline this week?	Unanswerable	None	On-call rotates [1]; the specific person is not mentioned.	Correct abstention	OK. No name was invented.

Result count

Result	Count
Answerable questions, grounded	8 of 10
Answerable questions, partially grounded	2 of 10 (A08, A10)
Unanswerable questions, correct abstention	3 of 3
Hallucinations (invented answers)	0

Part 4 · Failure analysis

A RAG system has two main parts. **Retrieval** finds the chunks that match the question. **Generation** is the model writing the answer with those chunks. For each problem below, the analysis says where the problem happened and what could fix it.

A08 · Broken number and repeated sentences

What happened	The answer says the pipeline fails when quarantined rows exceed "2". The % sign is missing, a citation number is placed inside the number, and two sentences appear twice. It also does not say "from HIGH rules" or "of the batch".
Where	Generation. The answer uses the correct facts from quality_rules.md (the quarantine threshold, the quarantine location and the failed_rule column), so retrieval found the right chunk.
Why	Nova Micro is a very small model. With several chunks that say similar things, it mixed and repeated the text when it added citations.
Possible fix	Use a larger approved model, or add a custom prompt that asks for one short sentence with the exact number and unit.

A10 · The alert channel is not named

What happened	The first part (quarantine location and DQ-004) is correct. For the second part, the model only repeats that the channel is "described in the pipeline README". It does not say SNS topic sns-orders-pipeline-alerts, Slack #data-eng-alerts or the team email.
Where	Most likely retrieval. This is a multi-document question. The chunk from quality_rules.md that points to the README was found, and a README chunk about on-call was found, but the "Retries and alerts" section of the README does not seem to be in the answer.
Why	Only 5 chunks are used per question. The question talks about currency and quarantine, so most of the 5 chunks came from quality_rules.md. The alerts chunk ranked lower.
Possible fix	Increase the number of source chunks (for example to 8 or 10), or ask two separate questions. An agentic RAG approach would split the question into two searches automatically.

U01 · Correct abstention with extra information

What happened	The model correctly says it cannot find the cost. But then it lists AWS pricing pages with links.
Where	Generation. The links are not in any of the four documents.
Why	The default prompt lets the model add helpful general knowledge after it says it does not know.
Possible fix	Add a custom prompt that says: answer only with the search results, and if the answer is not there, reply with one fixed sentence and nothing else.